Engine Model(s): PT6A

72-00-00

ENGINE, TURBOPROP - REMOVAL/INSTALLATION

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire, unless otherwise specified, shall comply with specification AMS5687, heat and corrosion resistant steel wire MS9226-03, which is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC09-005	Compound, Universal
PWC09-006	Compound, Universal

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name	Application
PWC30037	DELETED	
PWC30269	Mount Ring	
PWC34300	Stand, Engine	
PWC34752	Adapter, Engine	Replaces PWC30712
PWC51861-600	Sling Assembly, Engine	
PWC70099	DELETED	
PWC71296	Sling, Power Section	
PWC77193	Sling, Power Section	(Without Propeller)

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Removal/Installation of Engine

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A. Removal of Engine from Airframe

CAUTION: IF ENGINE IS DROPPED, IN CONTAINER OR OTHERWISE, SEND THE ENGINE TO AN APPROVED OVERHAUL FACILITY FOR LIGHT OVERHAUL. INDICATE "DROPPED ENGINE".

(1) Remove propeller and engine-mounted airframe-supplied accessories from engine (Ref. Aircraft Maintenance Manual).

CAUTION: THE AIRFRAME MOUNTED IGNITION EXCITER MUST BE REMOVED WITH THE ENGINE AND NOT LEFT CONNECTED TO AIRFRAME INPUT LEADS.

(2) Use sling (PWC51861-600) and suitable hoist to remove engine from airframe (Ref. Aircraft Maintenance Manual).

NOTE: After removal of engine from airframe, install correctly in shipping container if engine is to be shipped to an overhaul facility, or in engine stand if heavy maintenance is to be done (Ref. SERVICING).

B. Installation of Engine in Airframe

(1) Install engine-mounted airframe-supplied accessories (Ref. Aircraft Maintenance Manual).

(2) Use sling (PWC51861-600) and suitable hoist to install engine on airframe (Ref. Aircraft Maintenance Manual).

NOTE: Where ignition exciter is airframe mounted, install unit in accordance with Aircraft Maintenance Manual.

(3) Install propeller and other airframe-supplied, engine-mounted accessories on engine (Ref. Aircraft Maintenance Manual).

6. Removal/Installation of Power Section (P/S) from/to Gas Generator Assembly

CAUTION: IF P/S IS DROPPED, IN CONTAINER OR OTHERWISE, SEND THE PS/ENGINE TO AN APPROVED OVERHAUL FACILITY FOR LIGHT OVERHAUL. INDICATE "DROPPED ENGINE".

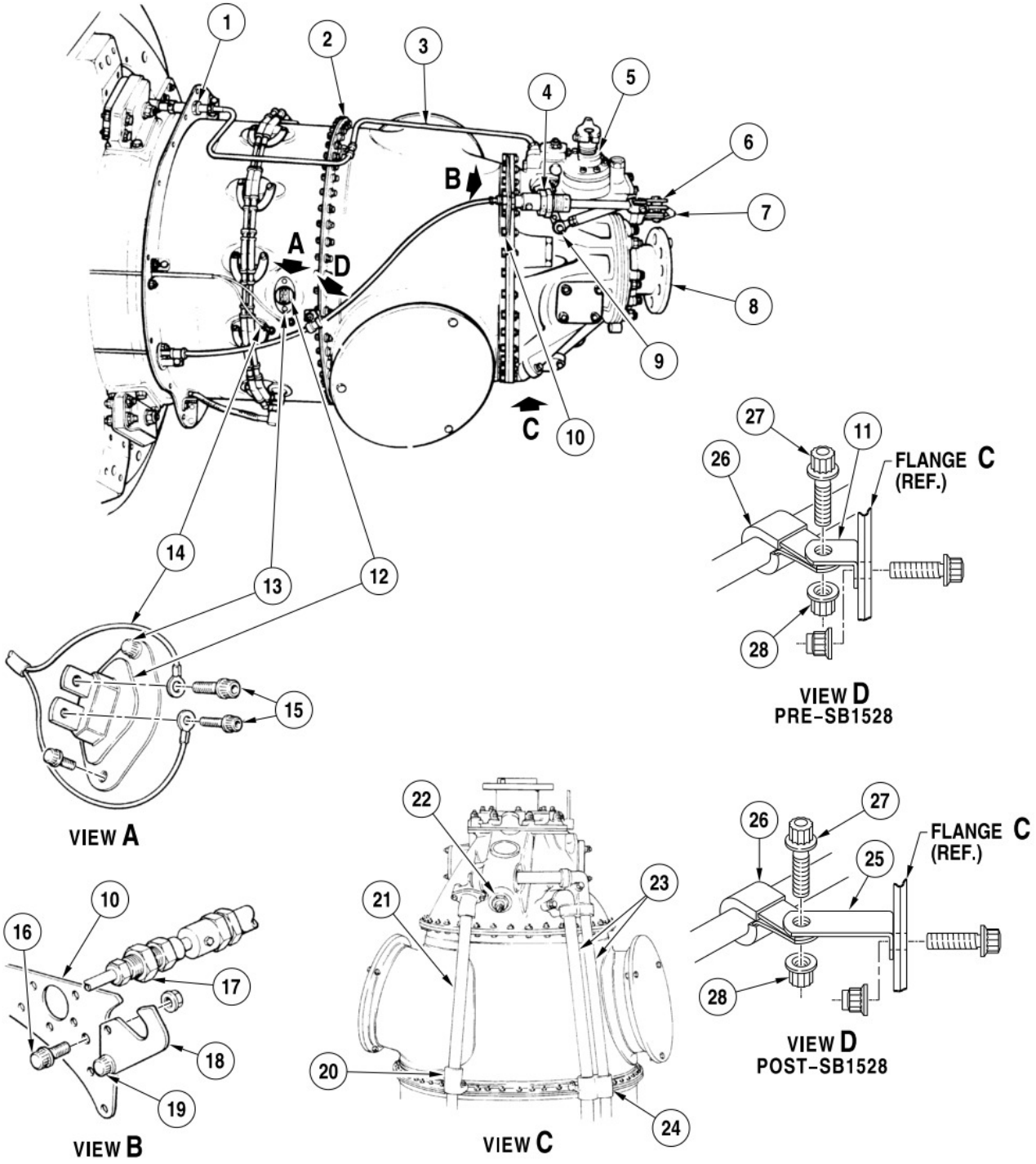
A. Removal

CAUTION: P/S MUST NOT BE LEFT IN SLING OVERNIGHT OR FOR LONGER PERIOD. REMOVE PROPELLER AND INSTALL P/S IN STAND (REF. PARA. 7.).

NOTE: Removal of power section from airframe mounted engine may be done with propeller installed or removed.

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Figure 401 Removal/Installation of Power Section



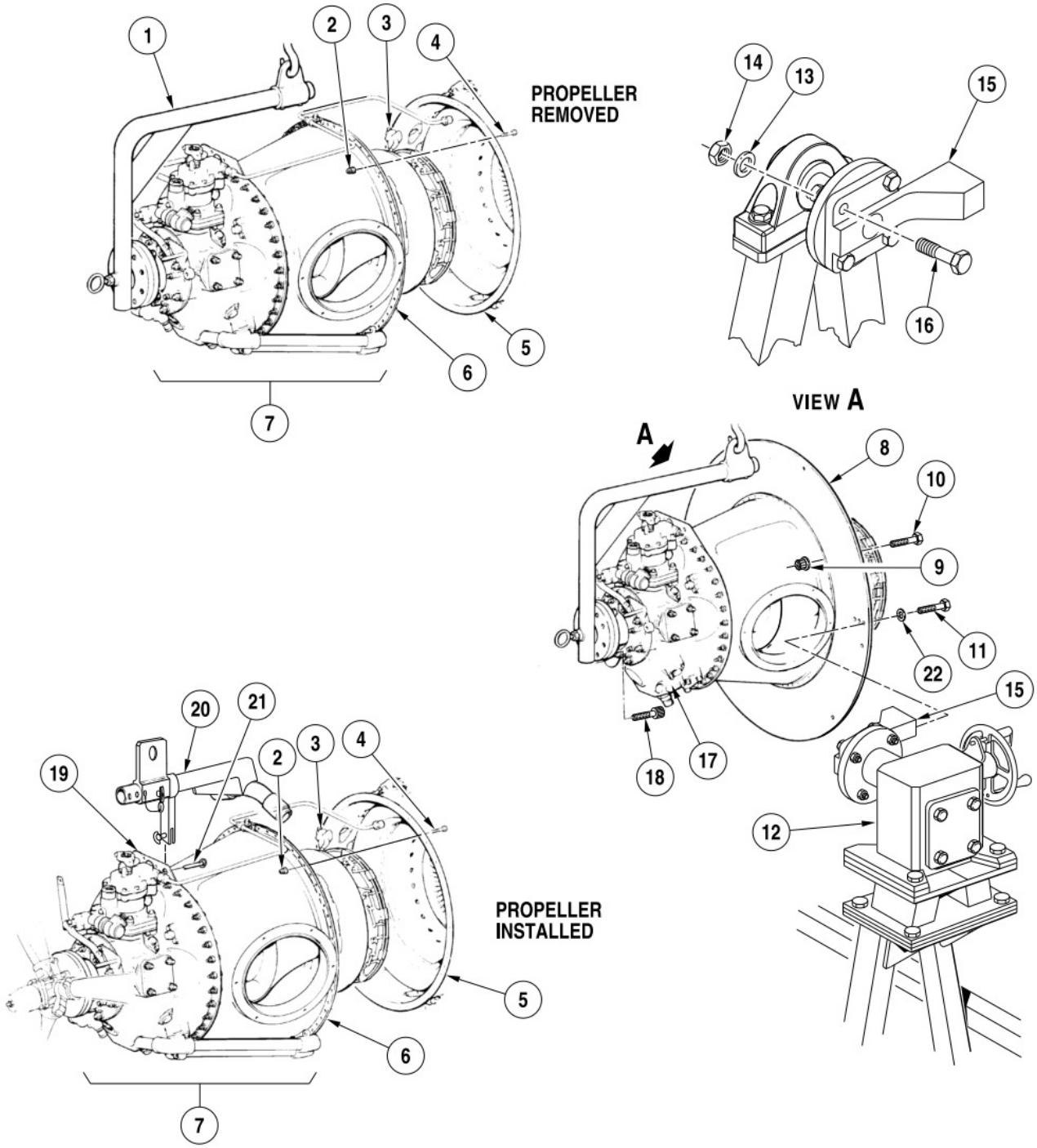
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1. Coupling Nut
2. Flange C
3. Propeller Governor Pnuematic (Py) Line
4. Propeller Governor Reset Arm
5. Propeller Governor
6. Clevis
7. Propeller Reversing Lever
8. Propeller Shaft Flange
9. Interconnect Rod
10. Front Lifting Bracket
11. Support Bracket (Pre-SB1528)
12. T5 Terminal Block
13. Retaining Bolt
14. Trim Harness Leads
15. Alumel and Chromel Bolts
16. Bolt
17. Locknut
18. Swivel Joint Retaining Plate
19. Bolt
20 Pressure Oil Tube Coupling
21. Pressure Oil Tube
22. Chip Detector
23. Scavenge Oil Tubes
24. Scavenge Oil Tube Coupling
25. Support Bracket (Post-SB1528)
26. Loop Clamp
27. Bolt
28. Nut

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Figure 402 Power Section Removal from Engine Gas Generator/Stand



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1. Sling Assembly (propeller not installed) (PWC77193)
2. Nut
3. T5 Terminal Block
4. Bolt, Tee Head
5. Gas Generator Assembly
6. Flange C
7. Power Section
8. Mount Ring (PWC30269)
9. Nut
10. Bolt
11. Screw
12. Stand (PWC34300)
13. Washer
14. Nut
15. Adapter
16. Bolt
17. Propeller Reduction Gearbox
18. Capscrew
19. Front Lifting Bracket
20. Sling Assembly (propeller installed) (PWC71296)
21. Ball Lock Detent Pin
22. Washer

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(1) Place suitable container under reduction gearbox and remove chip detector (22, Fig. 401) from boss at 6 o'clock position on front case of reduction gearbox. Allow oil to drain (Ref. SERVICING).

(2) Disconnect coupling nut (1) of governor pneumatic (Py) line (3) from center fireseal bulkhead coupling tube. Cap line and coupling tube.

(3) Disconnect electrical connections from airframe and engine accessories mounted on reduction gearbox, as necessary.

(4) For engines equipped for propeller reversing:

(a) Remove the cotterpin, nut, bolt and washer that attach the clevis end (6) of the push-pull terminal to the propeller reversing lever (7).

(b) Remove the cotterpin, nut, washers, spacer and bolt that attach the propeller governor interconnecting rod end (9) to the airbleed link reset arm (4) on the propeller governor (Ref. 76-10-00).

(c) Release locknut (17) that attaches the front swivel joint of the propeller reversing linkage to the front lifting bracket (10) (Ref. 76-10-00).

(d) Remove the inner nut and bolt (16) and slacken outer nut and bolt (19) that attach the swivel joint retaining plate (18) to the front lifting bracket (10). Swing away the retaining plate and lift the forward end of propeller reversing linkage clear of the front lifting bracket. Suitably support linkage (Ref. 76-10-00).

(e) Remove the nut (28) and bolt (27) that attach the loop clamp (26) and the support bracket (11, Pre-SB1528) or (25, Post-SB1528) at flange C (Ref. View D).

(f) Remove the nut and bolt that attach the bracket (11, Pre-SB1528) or (25, Post-SB1528) on flange C. Remove the bracket (11, Pre-SB1528) or (25, Post-SB1528) (Ref. View D).

(5) Remove alumel and chromel bolts (15) that attach T5 trim wiring harness (14) to T5 thermocouple terminal (12) on gas generator case. Move wiring harness lugs away from terminal. Temporarily attach T5 trim harness, and airframe leads if applicable, to gas generator case. Install alumel and chromel bolts again in terminal and tighten with fingers.

(6) Remove two bolts (13) that attach T5 wiring harness terminal (12) to gas generator case, then push terminal and gasket inside case.

CAUTION: WHEN POWER SECTION IS REMOVED FROM AIRFRAME MOUNTED ENGINE FOR OVERNIGHT, OR LONGER PERIODS OF TIME, THE PROPELLER MUST BE REMOVED (REF. AIRCRAFT MAINTENANCE MANUAL) AND POWER SECTION INSTALLED IN MAINTENANCE STAND (REF. PARA. 7.).

(7) Attach appropriate sling to power section:

(a) When propeller is not installed, position sling (PWC77193) (1, Fig. 402) on propeller shaft flange, engage with dowels and attach with four capscrews (18). Tighten the screws.

(b) When propeller is installed, position sling (20) (PWC71296) and attach to front lifting bracket (19) with pin (21).

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(8) Position hoist hook into sling (PWC77193) (1) or (PWC71296) (20) eye and take up slack on hoist.

CAUTION: DO NOT TURN OR DISTURB THE INTERSTAGE SEALING RING(S) ON THE POWER TURBINE SHROUD HOUSING, UNLESS THEY ARE TO BE REMOVED FOR INSPECTION (REF. 72-50-03).

(9) Remove nuts and bolts at Flange C (2, Fig. 401) and separate power section from the gas generator section. Using sling, partially withdraw power section. Retrieve and discard T5 wiring harness terminal gasket. Install the retaining bolts (13) again in T5 terminal block (12) and tighten the bolts with your hand.

(10) Cap tubes (21 and 23, Fig. 402) and install plugs on couplings (20 and 24).

(11) Withdraw power section from gas generator case.

B. Installation

NOTE: Whenever a replacement power section is installed on an engine at maintenance, the original power turbine stator assembly should be retained and installed on new power section whenever possible. This is to maintain vane throat area match between compressor turbine and power turbine vane rings, which can affect operating parameters of the engine if altered.

(1) Remove caps from pressure and scavenge oil transfer tubes on gas generator case and install new preformed packing on tubes.

(2) Remove plugs from coupling at Flange C on power section.

(3) Inspect mating areas of gas generator case and power section for cleanliness and security of parts; remove any protective covers installed during maintenance period.

NOTE: If required, a light coating of sealant (PWC09-005) or (PWC09-006) may be applied to mating faces of Flange C and to T5 terminal block and boss prior to assembly.

(4) Attach new gasket to T5 terminal block (3, Fig. 402) and insert through gas generator case. Attach terminal block with two retaining bolts (13, Fig. 401); tighten bolts to snug fit. Do not torque at this time.

(5) Prior to mating gas generator case to the power section, centralize the interstage sealing ring(s) (Ref. 72-50-03).

CAUTION: MAKE SURE FLANGE C OF GAS GENERATOR CASE AND EXHAUST DUCT ARE PARALLEL AND AXIALLY ALIGNED. IF ASSEMBLIES DO NOT MATE FREELY, DO NOT USE FORCE. WITHDRAW POWER SECTION AND CHECK FOR CAUSE OF INTERFERENCE.

(6) Move power section carefully into position, align pressure and scavenge oil transfer couplings (Ref. Fig. 403) with tubes at gas generator case, and mate power section with gas generator.

(7) Install the support bracket (11, Pre-SB1528, Fig. 401) or (25, Post-SB1528) on the flange C at location shown in Fig. 403 with a bolt (4, Fig. 402) (bolt head at rear) and nut (2). Tighten the

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nut (2) with your fingers.

NOTE: For Post-SB1528 engines, install the support bracket (25) with short leg on the flange C.

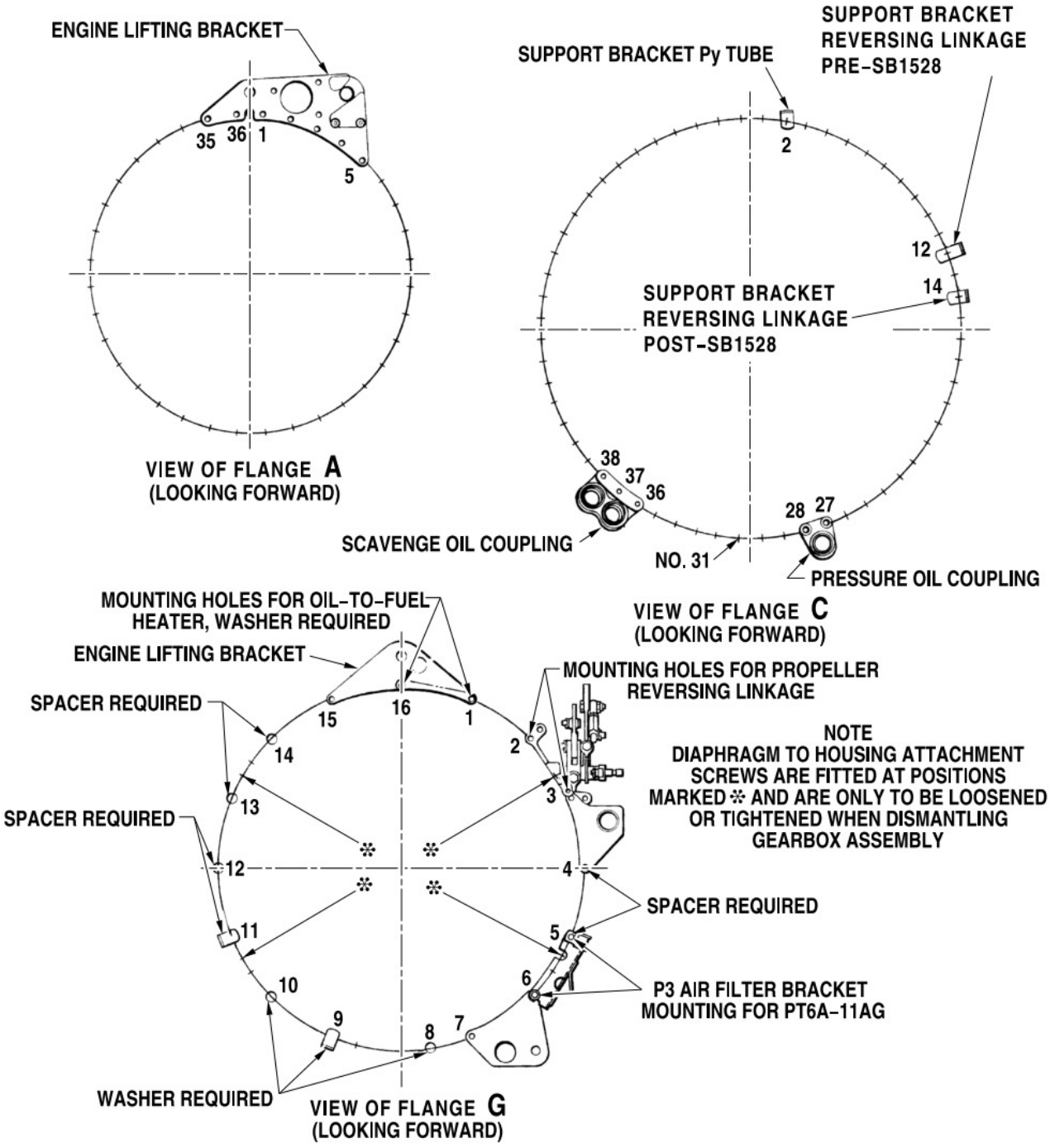
(8) Install the bolts (4, Fig. 402) through the remaining locations of the flange C with the bolt heads at rear and attach with the self-locking nuts (2).

(9) Torque the nuts (2, Fig. 402) in diametric sequence 36 to 40 lb.in.

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Figure 403 Bolting and Component Configuration

(SHEET 1 OF 2)

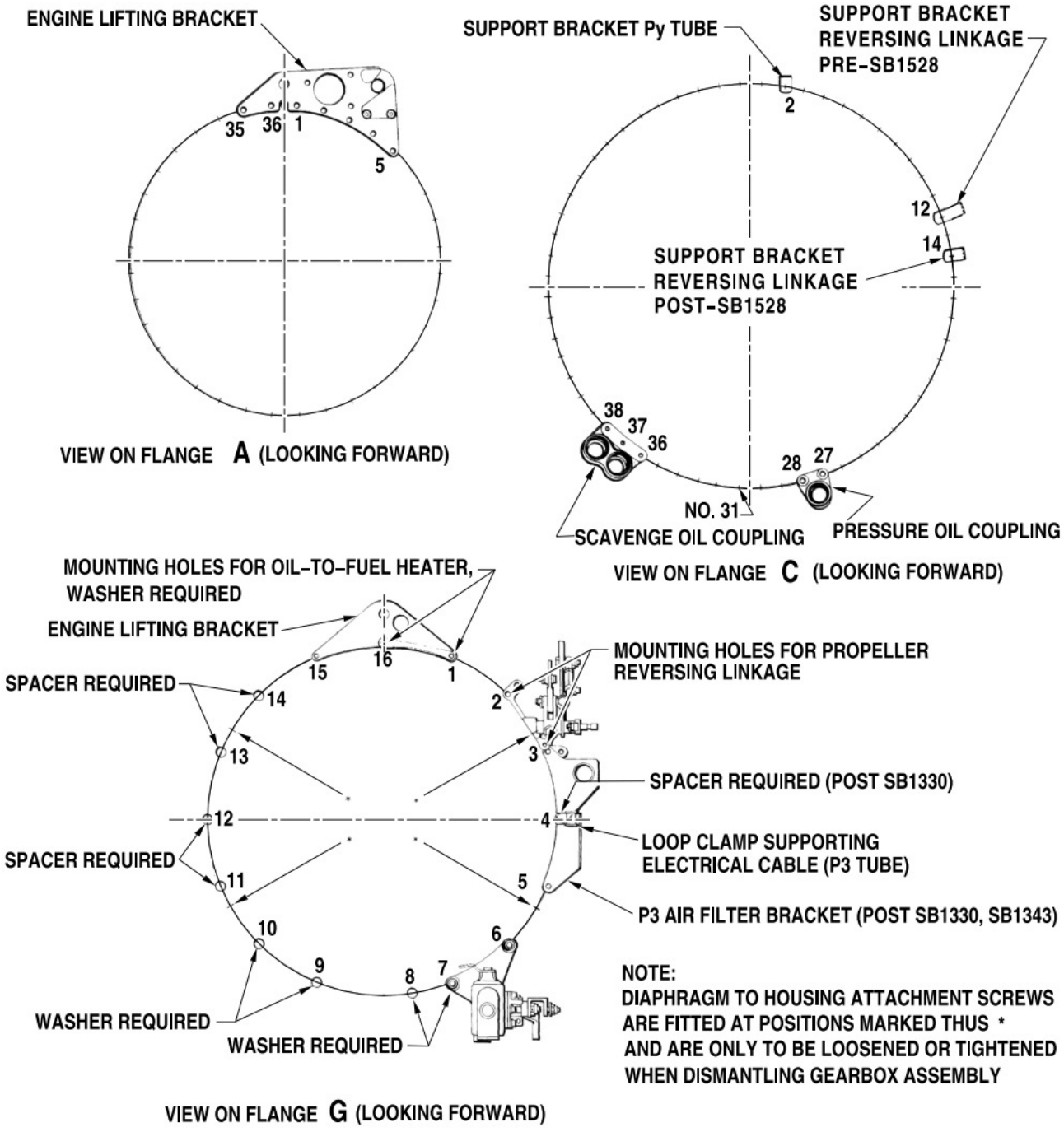


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PT6A-27 and PT6A-28

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(10) Remove the sling (PWC77193) (1) or (PWC71296) (20) from the power section (7).

(11) Torque terminal block mounting bolts (13, Fig. 401) 32 to 36 lbf.in. Do not lockwire until engine ground checks are completed.

(12) Connect leads (14, Fig. 401) of T5 trim harness and airframe leads to terminal block (12). Attach alumel terminal leads with large bolt, tighten 10 to 15 lb.in. Attach chromel terminal leads with small bolt, tighten 8 to 12 lb.in. (Ref. 77-20-01).

(13) Install new preformed packing on the chip detector (22) and install the detector in the reduction gearbox drain boss. Torque the detector 45 to 55 lbf.in. Locate cover on the chip detector and tighten with your hand. Attach the cover and detector to the reduction gearbox with lockwire.

(14) Install, tighten and lockwire airframe electrical fittings, as appropriate (Ref. Aircraft Maintenance Manual).

(15) Connect propeller governor pneumatic (Py) line coupling nut (1, Fig. 401) to nipple fitting on propeller governor. Torque nut 90 to 100 lb.in., and safety wire.

(16) For engines equipped with the propeller reversing mechanism:

(a) Assemble the loop clamp (26, Fig. 401) on the casing and attach to the angle bracket (11, Pre-SB1528) or (25, Post-SB1528) on flange C with a bolt (27) and nut (28). Torque the nut (28) 36 to 40 lb.in. (Ref. View D).

(b) Install the front swivel joint of the propeller reversing linkage in slot of the front lifting bracket (10, Fig. 401). Swing the retaining plate (18) over the swivel joint and install the inner nut and bolt (16), torque 36 to 40 lb.in. Tighten the front swivel joint locknut (17), torque 200 to 225 lb.in., and lockwire.

(c) Install the spacer sleeve in upper hole of the propeller reversing lever (7, Fig. 401). Locate the clevis end (6) of the push-pull terminal over upper end of the lever, align holes and insert the bolt. Attach with the castellated nut and washers, tighten the nut, torque 24 to 36 lb.in., and install the cotterpin.

(d) Locate lower end of the governor interconnecting rod (9, Fig. 401) on the propeller governor reset arm and Attach with a bolt, washer and castellated nut. Tighten the nut, torque 24 to 36 lb.in., and install the cotterpin.

(17) Do engine ground run checks (Ref. 71-00-00, ADJUSTMENT/TEST).

7. Removal/Installation of Power Section (P/S) from/in Maintenance Stand

A. Removal

(1) Position sling (PWC77193) (1, Fig. 402) on propeller shaft flange, engage with dowels and attach with four capscrews (18). Tighten the screws.

(2) With power section directly beneath hoist, engage hoist hook in sling eye and take up slack on hoist.

(3) Remove 12 nuts (9) and bolts (10) attaching power section to mount ring (8).

(4) Carefully remove the power section (7) from the mount ring (PWC30269) (8).

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(5) Install propeller governor pneumatic (Py) line (3, Fig. 401) to propeller governor. Torque coupling nut 90 to 100 lb.in.

(6) Install new preformed packings on pressure oil tube (21, Fig. 401) and scavenge oil tubes (23). Install tubes in respective bosses on reduction gearbox (17) and install couplings (20 and 24) on rear ends of tubes.

B. Installation

(1) Assemble adapters (15, (PWC34752), Fig. 402) to stand (12, PWC34300) with four bolts (16), washers (13) and nuts (14). Tighten the nuts.

(2) Remove propeller governor pneumatic (Py) line (3, Fig. 401), scavenge oil tubes (23) and pressure oil tube (21) from power section.

(3) Remove pressure oil tube (21) and scavenge oil tubes (23). Remove tubes in respective bosses on reduction gearbox (17) and couplings (20 and 24) on rear ends of the tubes.

(4) Position the mount ring (PWC30269) (8, Fig. 402) in stand (PWC34300) (12) with front face against rear face of adapters. Align and attach mount ring (8) to bracket with three washers (22) and screws (11). Tighten the screws.

(5) Lower power section to align Flange C (6, Fig. 402) with cutout in mount ring (8). Carefully position power section so that Flange C is in contact with mount ring.

(6) Attach power section (7) to mount ring (8) with 12 nuts (9) and bolts (10).

(7) Lower hoist to transfer power section weight to stand (PWC34300) (12). Remove sling (PWC77193) (1) from propeller shaft flange.

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